

Surgical Correction of Transposition of the Great Vessels

Technical Considerations

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ABSTRACT Intraatrial transposing of the venous return is a satisfactory method of treating transposition of the great arteries. Serious complications include atrioventricular conduction disturbances and pulmonary venous obstruction. Simple technical modifications that should eliminate these complications are described.

The intraatrial transposing of the venous return has been a highly effective method of correction in transposition of the great arteries [1, 4]. Although systemic venous obstruction has been reported [3], the most serious complication is pulmonary venous obstruction produced by the projection of the baffle across the pulmonary veins and into the right atrium. Tailoring [2] the pericardial baffle or enlarging [4, 5] the right atrium by means of a gusset have been suggested as methods of preventing this complication. Tailoring of the baffle is time-consuming, and if the pericardial patch is overly narrowed in the center, interference with the systemic venous return may result. Insertion of a gusset in the right atrial wall for the prevention of pulmonary venous obstruction is also a lengthy procedure; since the enlarging patch must be placed in the thin portion of the wall, excessive bleeding may be the ensuing complication.

We have used a method in the past few years which we believe prevents obstruction of pulmonary venous flow, does not significantly increase operative time, and does not add to the risk of bleeding. With this method, there has been no systemic venous obstruction or interference with

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atrioventricular conduction. After the mediastinum is opened through a sternum-splitting incision, the pericardial patch for the baffle is excised as a rectangle extending from phrenic nerve to phrenic nerve and from the level of the atrial appendages to the apex of the ventricles. This provides a large piece of pericardium, all of which will be used with no additional tailoring. The remaining portion of pericardium is opened with a vertical incision to the arch of the aorta and to the diaphragm.

Perfusion is through the ascending aorta. Venous return is accomplished by placing one catheter through the tip of the right atrial appendage into the superior vena cava and the other catheter through the right atrial wall at the junction of the inferior vena cava and the right atrium. After cardiopulmonary bypass has been instituted and the right atrium has been opened with a longitudinal incision only slightly longer than the extent of the interatrial septum, the entire remnant of the interatrial septum is removed, leaving only a small triangular ledge extending upward from the coronary sinus to a point on the interventricular septum one-third of the way down from its upper end. This remnant ledge protects the bundle of His (Fig. 1). Any holes produced in the heart by removal of the septum are repaired with figure-of-eight sutures. The pericardial baffle is then sutured into position. A continuous suture is started at a point midway between the left upper and left lower pulmonary veins. The suture passes around the margin of the left superior pulmonary vein, then across the atrium to the upper margin of the right superior pulmonary vein. From here it is continued to the wall of the right atrium and around the inferior margin of the right atrial appendage. The suture line is brought back to the interventricular septum, where it is temporarily interrupted. Another continuous suture is started from the original point, carried around the inferior margin of the left lower pulmonary vein, and continued to the inferior margin of the right lower pulmonary vein to the free right atrial wall and around the margin of the inferior vena cava to the interventricular septum. The suture at this point is well below the coronary sinus and

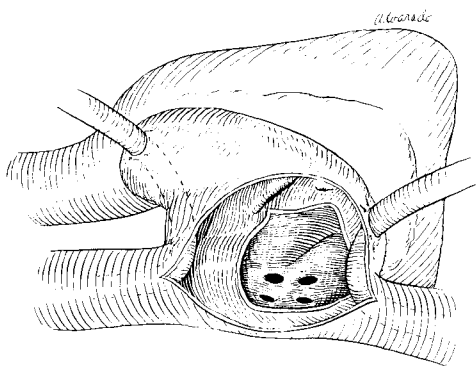
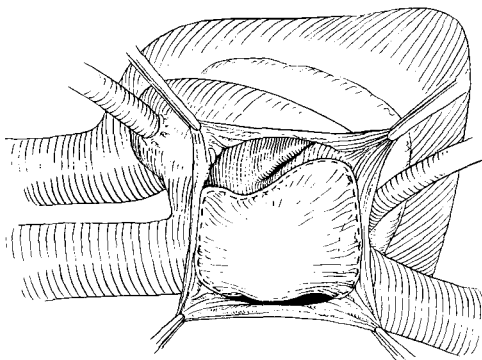


FIG. 1. The entire interatrial septum is completely removed except for a small rim from the coronary sinus to the upper portion of the interventricular septum.

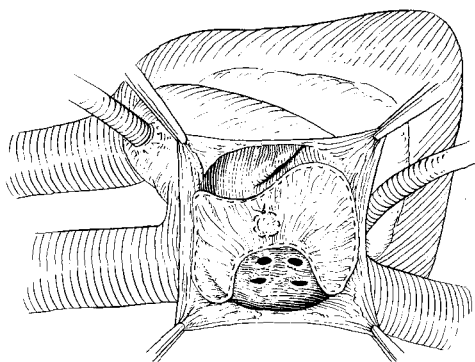
FIG. 2. The pericardial baffle is sewn to the base of the tricuspid leaflets rather than to the cut margin of atrial septum. The patch balloons into the path of pulmonary venous flow and can cause obstruction.



atrioventricular node. It is continued on to the base of the tricuspid valve in such a manner that the pericardium is sewed to the tricuspid leaflets at their base rather than into the septum. The suture line is continued upward to the midpart of the septal leaflet. The upper suture is brought to this point, and both sutures are tied. Note that the suture line here is placed in the base of the tricuspid leaflet, in much the same manner as when an ostium primum defect is repaired, instead of in the cut margin of the septum.

Following complete transposition of the venous return with this large pericardial graft, there is ample room for caval flow into the left atrium; but the ballooning of the pericardium tends to block the pulmonary venous flow into the right atrium (Fig. 2). This is particularly noticeable if the inferior caval tourniquet is released and that caval catheter momentarily clamped. The pericardium is then depressed in the center until the pulmonary veins can be visualized without difficulty; a purse-string suture is placed in the pericardium, centered on the point of maximal depression. When this suture is tied with its center inverted to the left, there is no longer an obstruction to pulmonary venous flow (Fig. 3). The resulting indentation does not obstruct caval flow but seems to divert it almost straight to the left. As the right atrium is closed from each end toward the center, one can see that there is sufficient room for pulmonary venous return around the baffle to the

FIG. 3. A purse-string suture has been placed in the pericardial patch so that the pulmonary veins can no longer be obstructed.



tricuspid valve. Removal of air and termination of cardiopulmonary bypass are done by the standard procedures.

Three points in the above description should be noted: (1) The small ledge of interatrial septum from the coronary sinus to the ventricular septum is left in order to eliminate any danger of injury to the atrioventricular node or bundle of His. (2) The pericardial patch is sutured to the base of the tricuspid leaflets rather than to the cut margin of the septum and the rim of the coronary sinus. This may give more room for systemic venous flow to the left, and it eliminates the danger of producing atrioventricular conduction difficulties. (3) The indentation of the center of the patch with purse-string or other sutures allows free flow of the pulmonary venous return into the right atrium and tricuspid valve without operative enlargement of the right atrium.

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